
PROFILE

Computer vision engineer with a geoinformatics background. I train detection models on mobile mapping imagery, orthophotos and lidar point clouds, and solve what comes next: lifting detections into real space without precise GNSS — multi-view triangulation, least-squares refinement, monocular depth estimation and cross-validation against a LiDAR point cloud. Production R&D experience at CEDA Maps, research presented at the POGEO 2026 conference.

EXPERIENCE

GIS & Computer Vision Analyst (contractor) — CEDA Maps

June 2026 – present

Brno, remote

- Building a GeoAI pipeline for automatic detection and classification of road signs from mobile mapping — from data annotation through model training to production deployment
- Localising signs without precise GNSS: multi-view triangulation of observations from dozens of passes, least-squares refinement and fusion into a single entity; monocular distance estimation (Depth Anything V2, MoGe-2) as an independent source, validated by a null test and scenario synthetic data
- Calibration of a panoramic camera rig via COLMAP SfM — extrinsics and intrinsics for lifting detections into real-world space
- Detecting sign panels directly in the LiDAR point cloud (RIEGL) and cross-validating them against image detections
- Dataset engineering: per-track splits against data leakage, deduplication of sequential frames and class-imbalance control over tens of thousands of crops; synthetic scene augmentation for weather and lighting
- Python, PyTorch, Ultralytics YOLOv8/OBB, SAHI, ONNX Runtime, OpenCV, Open3D, laspy, COLMAP, CVAT, PostGIS, GeoPandas, Shapely

Co-founder and technical lead — VečerkaPlus

April 2026 – present

Frýdek-Místek, Czechia · vecerkaplus.cz

- Co-owner and technical lead of a late-night drinks and snacks delivery service in Frýdek-Místek — the entire web platform built from scratch on React, Supabase and Vercel
- Delivery zone system built on spatial analysis: 1,093 grid points via the Google Distance Matrix API, so price and availability follow real driving distance rather than straight-line distance
- Alongside development I run operations — purchasing, courier logistics and growth strategy; average revenue of CZK 452 per order at a 36.5 % margin

Web Developer & Administrator — Palacký University Olomouc

May 2023 – present

Olomouc, Czechia

- Running olomouckymajales.cz and meetup.upol.cz — 199,800 visitors and 452,000 page views in total
- Requested changes delivered within 24 hours and uninterrupted operation during key events; design and functional changes deployed to production with no downtime

GIS & Remote Sensing Analyst — Skymaps s.r.o.

February 2025 – September 2025

Czechia

- Processing multispectral satellite imagery into thematic maps of soil potential
- Analysis and interpretation of remote sensing data: Sentinel in SNAP, GDAL, EO Browser; Python automation from batch image processing to final map outputs

Drone Operator — Aerial imaging (freelance)

2024 – 2025

Ostrava, remote

- Commissioned drone footage for corporate marketing materials; planning and flying missions in compliance with EASA A1/A3 rules (DJI Mini 3 Pro)

SELECTED PROJECTS

Vehicle Detection on Orthophotos

2025 – 2026

github.com/MetrPikeska/vehicle-detection-ortophoto

Vehicle detection on large-area orthophotos of Olomouc, fully integrated into a GIS workflow. YOLOv8-OBB with SAHI enables inference on images larger than the GPU can hold; DBSCAN clustering of detections and Voronoi-based address assignment in GeoPandas follow. An end-to-end pipeline from CVAT annotation through training and georeferencing to spatial analysis. Presented at the international POGEO 2026 conference.

Python · YOLOv8-OBB · SAHI · GeoPandas · Shapely · CUDA

Parking Occupancy Detection from a Camera Stream

2025 – 2026

A system deriving parking occupancy from site cameras that runs unattended: a scheduler triggers capture, detection and the write to PostgreSQL, with follow-up scripts backfilling gaps. Inference combines several detection models with custom overlap merging, and frames are georeferenced into the Czech national grid. A separate branch reads number plates and validates the OCR output against a Czech plate mask, so nonsense never reaches the data. The dashboard API is a deliberately separate FastAPI service, keeping GPU dependencies off the API server.

Python · Ultralytics YOLO · PyTorch · OpenCV · FastAPI · PostgreSQL · rasterio · pyproj · pytest

Roundabout Exit Detection

2025

github.com/MetrPikeska/01_Detekce_vyjezdu

A system for automatically counting vehicles leaving a roundabout. YOLOv8 detection with geometric tracking via Shapely polygons keeps IDs stable through occlusion; CUDA acceleration handles video sequences in real time even on a GTX 1060.

Python · YOLOv8 · OpenCV · Shapely · CUDA

SKILLS		
Languages & data-bases	Python · SQL / PL-pgSQL · JavaScript / TypeScript · R · Bash · C# · PostgreSQL / PostGIS · MySQL · Supabase	
Computer vision & machine learning	PyTorch · Ultralytics YOLOv8 / OBB · SAHI · OpenCV · torchvision · Hugging Face Transformers · Depth Anything V2 · MoGe-2 · scikit-learn · ONNX Runtime · OCR / ANPR · CVAT · CUDA	
GIS & spatial data	QGIS · ArcGIS Pro / ArcPy · GDAL/OGR · GeoServer · MapServer · SNAP · GeoPandas · Shapely · pyproj · Rasterio · H3 · Folium · OGC standards · EPSG 5514 / 3045 / 4326	
LiDAR, 3D & photogrammetry	Point cloud processing (RIEGL) · Open3D · laspy · CloudCompare · Agisoft Metashape · COLMAP (SfM) · multi-view triangulation · camera rig calibration · DTM/DSM · Blender · MeshLab	
Web & backend	React · Vite · Leaflet · FastAPI · Flask · Node.js / Express · REST API · WebSocket · HTML/CSS · WordPress · Vercel	
Tooling & operations	Git / GitHub · Linux / Ubuntu / WSL · Docker · pytest · NumPy · Pandas · SciPy · Matplotlib	
AI-assisted development	Claude Code — custom extensions, project rules and memory, orchestration of parallel agents · MCP clients (Supabase, Vercel, browser automation) · GitHub Copilot · AI code review in CI (GitHub Actions)	
Domain & standards	TP 65 (Czech road sign standard) · HD maps · mobile mapping systems · GNSS and NMEA · TomTom / StreetNet · RÚIAN · Sentinel / remote sensing · INSPIRE	
EDUCATION		
MSc — Geoinformatics and Cartography (in progress)		2025 – 2027
Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics		
– Focus: advanced spatial data processing, web cartography, remote sensing and image processing		
– Master’s thesis (in progress): <i>Evaluating LLM-generated maps against a traditional GIS workflow</i>		
– Erasmus+ — study period in Greece, October 2026 – February 2027		
– Drones, laser scanning, 3D printing and virtual reality		
BSc — Geoinformatics and Cartography		2022 – 2025
Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics		
– Bachelor’s thesis: <i>A user GIS toolbox for assessing the accessibility of parks and green spaces for urban residents</i> — grade B		
Maturita — Programming and Application Development / Internet of Things		2018 – 2022
Secondary School of Information Technology · Frýdek-Místek, Czechia		
– Programming, web technologies, computer networks, databases and the fundamentals of electronics and IoT		
CERTIFICATIONS		
October 2024	Remote pilot certificate, category A1/A3 · Civil Aviation Authority of the Czech Republic / EASA	
May 2023	Python for Everyone · Esri	
April 2026	ISSonVIS 2026 — International Spring School on Visualization · Dept. of Geoinformatics, Palacký University Olomouc	
LANGUAGES		
Czech (native) · English (B2)		